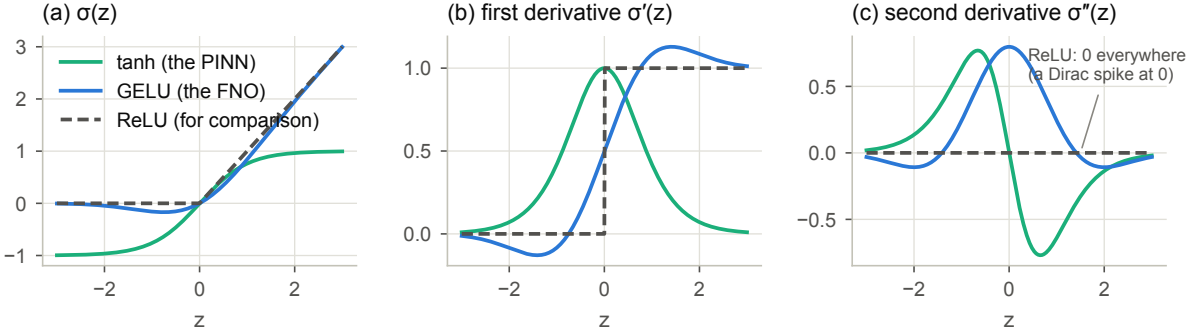
(a) σ(z)
tanh (the PINN)
GELU (the FNO)
ReLU (for comparison)
3
2
1
0
−1
−2
0
2
z
(b) first derivative σ′(z)
1.0
0.5
0.0
−2
0
2
z
(c) second derivative σ″(z)
ReLU: 0 everywhere
(a Dirac spike at 0)
0.5
0.0
−0.5
−2
0
2
z